

Blood is an opaque fluid with a viscosity greater than that of water (mean relative viscosity 4.75 at 18°C), and a specific gravity of 1.06 at 15°C. It is bright red when oxygenated, in the systemic arteries, and dark red to purple when deoxygenated, in systemic veins. Blood is a mixture of a clear liquid, plasma and cellular elements, and consequently the hydrodynamic flow of blood in vessels behaves in a complex manner that is not entirely predictable by simple Newtonian equations.

Plasma is a clear, yellowish fluid that contains many substances in solution or suspension: low-molecular-weight solutes give a mean freezing-point depression of 0.54°C. Plasma contains high concentrations of sodium and chloride ions, potassium, calcium, magnesium, phosphate, bicarbonate, traces of many other ions, glucose, amino acids and vitamins. It also includes high-molecular-weight plasma proteins, e.g. clotting factors, particularly prothrombin; immunoglobulins and complement proteins involved in immunological defence; glycoproteins, lipoproteins, polypeptide and steroid hormones, and globulins for the transport of hormones and iron. The plasma is involved in the transport of most molecules that are released or secreted by cells in response to pathological or physiological stimuli and so the routine chemical analysis of plasma is of great diagnostic importance. There is increasing interest in using metabolomics approaches for the high-throughput analysis of small molecules or metabolites in the serum, as a potential aid to diagnosis and understanding of disease (Psychogios et al 2011).

The precipitation of the protein fibrin from plasma to form a clot (Fig. 4.1) is initiated by the release of specific materials from damaged cells and blood platelets in the presence of calcium ions. If blood or plasma samples are allowed to stand, they will separate into a clot and a clear yellowish fluid, the serum. Clot formation is prevented by removal of calcium ions, e.g. by addition of citrate, oxalate or various organic calcium chelators (EDTA, EGTA) to the sample. Heparin is also widely used as an anticlotting agent because it interferes with fibrin clot formation.

In postnatal life, blood cells are formed in the bone marrow. Haemopoiesis produces red cells (erythrocytes) and a wide variety of

defensive cells (white blood cells, or leukocytes). The latter include neutrophil, eosinophil and basophil granulocytes, B lymphocytes and monocytes. T lymphocytes develop in the thymus from bone marrow-derived progenitors. These cells all contribute to the immune system of the human (for an overview of the immune system, see Murphy (2011)). Platelets are produced in the bone marrow as cellular fragments of megakaryocytes. Only erythrocytes and platelets are generally confined to the blood vascular system, whereas all leukocytes can leave the circulation and enter extravascular tissues. The numbers of cells doing so increases greatly during inflammation caused by local infections or tissue damage.

The lymphoid tissues are the thymus, lymph nodes, spleen and the lymphoid follicles associated mainly with the alimentary and respiratory tracts. Lymphocytes populate lymphoid tissues and are concerned with immune defence. Lymphoid tissue also contains supportive stromal cells, which are non-haemopoietic in origin (e.g. thymic epithelium); non-haemopoietic follicular dendritic cells of lymph nodes and splenic follicles; haemopoietically derived dendritic cells; and macrophages of the mononuclear phagocyte system. Dendritic cells and blood monocyte-derived macrophages are found additionally in most tissues and organs, where they function as antigen-presenting cells (APCs).

CELLS OF PERIPHERAL BLOOD

ERYTHROCYTES

Erythrocytes (red blood cells, RBCs) account for the largest proportion of blood cells (99% of the total number), with normal values of $4.1\text{--}6.0 \times 10^6/\mu\text{l}$ in adult males and $3.9\text{--}5.5 \times 10^6/\mu\text{l}$ in adult females. Polycythaemia (increased red cell mass) can occur in individuals living at high altitude, or pathologically in conditions resulting in arterial hypoxia. Reduction in red cell mass (anaemia) has many underlying causes but in rare instances can be due to structural defects in erythrocytes (see below).

Each erythrocyte is a biconcave disc (see Fig. 4.1; Fig. 4.2) with a mean diameter in dried smear preparations of 7.1 μm ; in fresh preparations the mean diameter is 7.8 μm , decreasing slightly with age. Mature erythrocytes lack nuclei. They are pale red by transmitted light, with

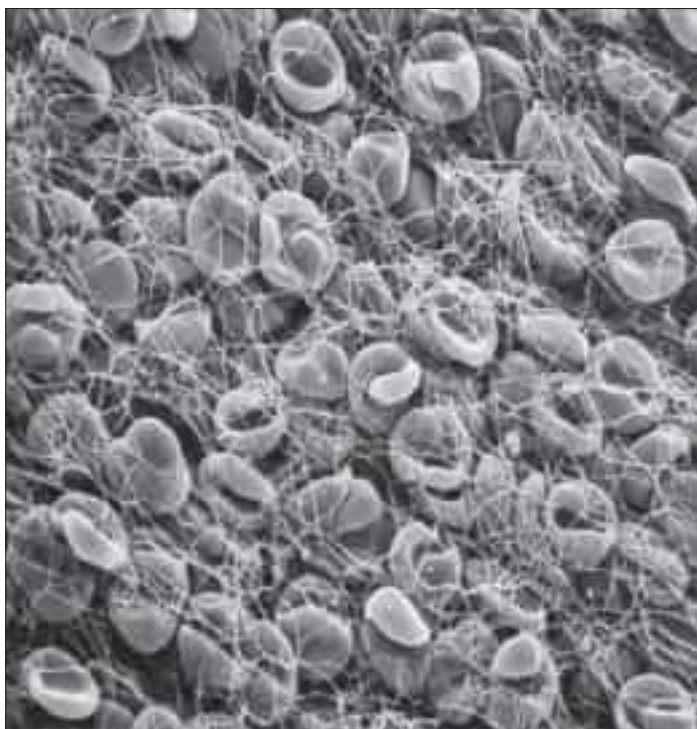


Fig. 4.1 Erythrocytes enmeshed in filaments of fibrin in a clot. (Courtesy of Michael Crowder MD.)

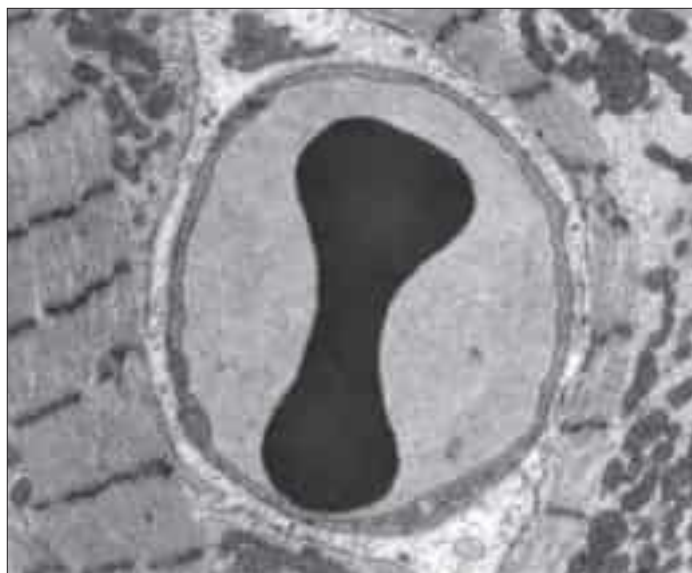


Fig. 4.2 A human heart muscle biopsy specimen, showing an erythrocyte within a capillary. The erythrocyte biconcave disc is typically electron-dense and almost fills the capillary lumen. (Courtesy of Dr Bart Wagner, Histopathology Department, Sheffield Teaching Hospitals, UK.)

paler centres because of their biconcave shape. The properties of their cell coat cause them to adhere to one another by their rims to form loose piles of cells (rouleaux). In normal blood, a few cells assume a shrunken, star-like, crenated form; this shape can be reproduced by placing normal biconcave erythrocytes in a hypertonic solution, which causes osmotic shrinkage. In hypotonic solutions erythrocytes take up water and become spherical; they may eventually lyse to release their haemoglobin (haemolysis), leaving red-cell ghosts.

Erythrocytes have a plasma membrane that encloses mainly a single protein, haemoglobin, as a 33% solution. The plasma membrane of erythrocytes is 60% lipid and glycolipid, and 40% protein and glycoprotein.

More than 15 classes of protein are present, including two major types. Glycophorins A and B (each with a molecular mass of approximately 50 kDa) span the membrane, and their negatively charged carbohydrate chains project from the outer surface of the cell. Their sialic acid groups confer most of the fixed charge on the cell surface. A second transmembrane macromolecule, band 3 protein, forms an important anion channel, exchanging bicarbonate for chloride ions across the membrane and allowing the release of CO₂ in the lungs.

The filamentous protein, spectrin, is responsible for maintaining the shape of the erythrocyte. A dimer is formed of $\alpha 1$ and $\beta 1$ spectrin monomers, and two dimers then come together to form a tetramer (Machnicka et al 2013). These are joined by junctional complexes that contain (among other proteins) ankyrin, short actin filaments, tropomyosin and protein 4.1, forming a hexagonal lattice that supports the plasma membrane (Mankelov et al 2012). The junctional complex also interacts with transmembrane proteins. This structure gives the membrane great flexibility; red cells are deformable but regain their biconcave shape and dimensions after passing through the smallest capillaries, which are 4 μm in diameter (Mohandas and Gallagher 2008). Erythrocyte membrane flexibility also contributes to the normally low viscosity of blood. Molecular defects in the cytoskeleton result in abnormalities of red cell shape, membrane fragility, premature destruction of erythrocytes in the spleen and haemolytic anaemia (Iolascon et al 1998).

Fetal erythrocytes up to the fourth month of gestation differ markedly from those of adults, in that they are larger, are nucleated and contain a different type of haemoglobin (HbF). After this time they are progressively replaced by the adult type of cell.

Haemoglobin

Haemoglobin (Hb) is a globular protein with a molecular mass of 67 kDa. It consists of globulin molecules bound to haem, an iron-containing porphyrin group. The oxygen-binding power of haemoglobin is provided by the iron atoms of the haem groups, and these are maintained in the ferrous (Fe²⁺) state by the presence of glutathione within the erythrocyte. The haemoglobin molecule is a tetramer, made up of four subunits, each a coiled polypeptide chain holding a single haem group.

Mutations in the haemoglobin chains can result in a range of pathologies (Forget and Bunn 2013).

Lifespan

Erythrocytes last between 100 and 120 days before being destroyed. As erythrocytes age, they become increasingly fragile, and their surface charges decrease as their content of negatively charged membrane glycoproteins diminishes. The lipid content of their membranes also reduces. Aged erythrocytes are taken up by the macrophages of the spleen (Mebius and Kraal 2005) and liver sinusoids, usually without prior lysis, and are hydrolysed in phagocytic vacuoles where the haemoglobin is split into its globulin and porphyrin moieties. Globulin is further degraded to amino acids, which pass into the general amino-acid pool. Iron is removed from the porphyrin ring and either transported in the circulation bound to transferrin and used in the synthesis of new haemoglobin in the bone marrow, or stored in the liver as ferritin or haemosiderin. The remainder of the haem group is converted in the liver to bilirubin and excreted in the bile. Haemoglobin that is released by destruction of erythrocytes in the body binds to haptoglobin, and is taken up via CD163 receptors expressed on the surface of macrophages (Kristiansen et al 2001).

The recognition of effete erythrocytes by macrophages appears to take place by a number of mechanisms (Bratosin et al 1998). These include the exposure of phospholipids (such as phosphatidyl serine) that are normally found on the inner leaflet of the membrane bilayer, alterations in the carbohydrates expressed by the cells (most notably the loss of sialic acid) and the binding of autoantibodies to antigens

exposed on the aged erythrocyte. These lead to the cells being recognised and taken up by macrophages. Red cells are destroyed at the rate of 5×10^{11} cells a day (or nearly 6 million a second) and are normally replaced from the bone marrow (see Fig. 4.12) at the same rate.

Blood groups

Over 300 red cell antigens are recognisable with specific antisera. They can interact with naturally occurring or induced antibodies in the plasma of recipients of an unmatched transfusion, causing agglutination and lysis of the erythrocytes. Erythrocytes of a single individual carry several different types of antigen, each type belonging to an antigenic system in which a number of alternative antigens are possible in different persons. So far, 19 major groups have been identified. They vary in their distribution frequencies between different populations, and include the ABO, Rhesus, MNS, Lutheran, Kell, Lewis, Duffy, Kidd, Diego, Cartwright, Colton, Sid, Scianna, Yt, Auberger, Ii, Xg, Indian and Dombrock systems. Clinically, the ABO and Rhesus groups are of most importance.

Leukocytes also bear highly polymorphic antigens encoded by allelic gene variants. These belong to the group of major histocompatibility complex (MHC) antigens, also termed human leukocyte antigens (HLA) in humans. HLA class I antigens are expressed by all nucleated cells. Class II antigens are expressed on antigen-presenting cells (APCs) of the immune system, but can also be induced on many parenchymal cell types, e.g. after exposure to interferon. HLA class I and II antigens play important roles in cell-cell interactions in the immune system, particularly in the presentation of antigens to T lymphocytes by APCs.

LEUKOCYTES

Leukocytes (white blood cells) belong to at least five different categories (see Fig. 4.12) and are distinguishable by their size, nuclear shape and cytoplasmic inclusions. In practice, leukocytes are often divided into two main groups: namely, those with prominent stainable cytoplasmic granules, the granulocytes, and those without.

Granulocytes

This group consists of eosinophil granulocytes, with granules that bind acidic dyes such as eosin; basophil granulocytes, with granules that bind basic dyes strongly; and neutrophil granulocytes, with granules that stain only weakly with either type of dye. Granulocytes (Fig. 4.3) all possess irregular or multilobed nuclei and belong to the myeloid series of blood cells (see Fig. 4.12).

Neutrophil granulocytes

Neutrophil granulocytes (neutrophils) are also referred to as polymorphonuclear leukocytes (polymorphs) because of their irregularly

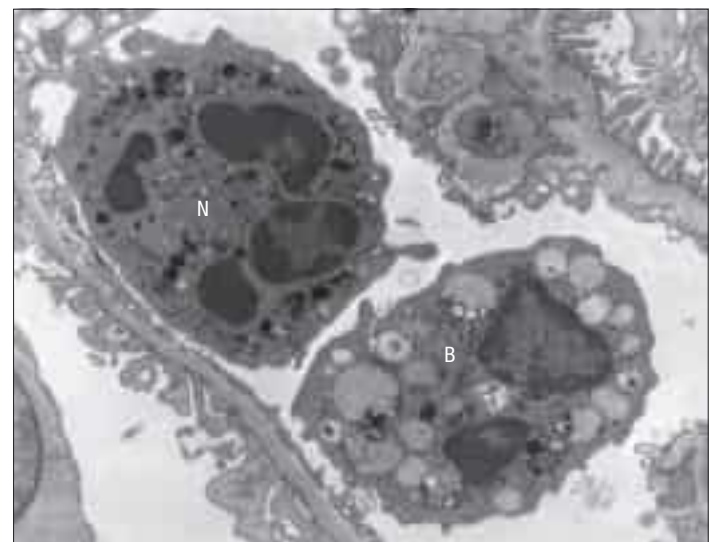


Fig. 4.3 Neutrophil (N) and basophil (B) granulocytes within a renal glomerular capillary in a human kidney biopsy. The neutrophil nucleus is more segmented (four lobes are visible) and the granules are smaller and more electron-dense than in the basophil. (Courtesy of Dr Bart Wagner, Histopathology Department, Sheffield Teaching Hospitals, UK.)

In normal blood, five types of polypeptide chain can occur: namely, α , β and two β -like polypeptides, γ and δ . A third, β -like η chain is restricted to early fetal development. Each haemoglobin molecule contains two α -chains and two others, so that several combinations, and hence a number of different types of haemoglobin molecule, are possible. For example, haemoglobin A (HbA), which is the major adult class, contains 2 α - and 2 β -chains; a variant, HbA₂ with 2 α - and 2 δ -chains, accounts for only 2% of adult haemoglobin. Haemoglobin F (HbF), found in fetal and early postnatal life, consists of 2 α - and 2 γ -chains. Adult red cells normally contain less than 1% of HbF.

In the genetic condition thalassaemia, only one type of chain is expressed normally, the mutant chain being absent or present at much reduced levels. Thus, a molecule may contain 4 α -chains (β -thalassaemia) or 4 β -chains (α -thalassaemia). In haemoglobin S (HbS) of sickle-cell disease, a point mutation in the β -chain gene (valine substituted for glutamine) causes the haemoglobin to polymerize under conditions of low oxygen concentration, thus deforming the red blood cell.

In the ABO system, two allelic genes are inherited in simple Mendelian fashion. Thus the genome may be homozygous and carry the AA complement, the blood group being A, or the BB complement, which gives blood group B, or it may carry neither (OO), producing blood group O. In the heterozygous condition the following combinations can occur: AB (blood group AB), AO (blood group A) and BO (blood group B). The ABO blood group antigens are all membrane glycolipids.

Individuals with group AB blood lack antibodies to both A and B antigens, and so can be transfused with blood of any group; they are termed universal recipients. Conversely, those with group O, universal donors, can give blood to any recipient, since anti-A and anti-B

antibodies in the donated blood are diluted to insignificant levels. Normally, however, blood is only transfused between persons with corresponding groups because anomalous antibodies of the ABO system are occasionally found in blood and may cause agglutination or lysis. The anti-ABO agglutinins, unlike those of the Rhesus system, belong to the immunoglobulin M (IgM) class and do not cross the placenta during pregnancy.

The Rhesus antigen system is determined by three sets of alleles: namely, Cc, Dd and Ee. The most important clinically is Dd. Inheritance of the Rh factor also obeys simple Mendelian laws; it is therefore possible for a Rhesus-negative mother to bear a Rhesus-positive child. Under these circumstances, fetal Rh antigens can stimulate the production of anti-Rh antibodies by the mother; as these belong to the IgG class of antibody they are able to cross the placenta. For most of the pregnancy the stroma stops the blood group antibodies from crossing into the fetal circulation. However, immediately prior to birth, the antibodies can cross this barrier and cause destruction of fetal erythrocytes. In the first such pregnancy little damage usually occurs because anti-Rh antibodies have not been induced, but in subsequent Rh-positive pregnancies massive destruction of fetal red cells may result, causing fetal or neonatal death (haemolytic disease of the newborn). Sensitization of the maternal immune system can also result from abortion or miscarriage, or occasionally even from amniocentesis, which may introduce fetal antigens into the maternal circulation. Treatment is by exchange transfusion of the neonate or, prophylactically, by giving Rh-immune (anti-D) serum to the mother after the first Rh-positive pregnancy, which destroys the fetal Rh antigen in her circulation before sensitization can occur.

segmented (multilobed) nuclei. They form the largest proportion of the white blood cells (40–75% in adults, with a normal count of 2500–7500/ μl) and have a diameter of 12–14 μm . The cells may be spherical in the circulation, but they can flatten and become actively motile within the extracellular matrix of connective tissues.

The numerous cytoplasmic granules are heterogeneous in size, shape and content, but all are membrane-bound and contain hydrolytic and other enzymes. Two major types can be distinguished according to their developmental origin and contents. Non-specific or primary (azurophilic) granules are formed early in neutrophil maturation. They are relatively large (0.5 μm) spheroidal lysosomes containing myeloperoxidase, acid phosphatase, elastase and several other enzymes. Specific or secondary granules are formed later, and occur in a wide range of shapes including spheres, ellipsoids and rods. These contain strong bacteriocidal components including alkaline phosphatase, lactoferrin and collagenase, none of which is found in primary granules. Conversely, secondary granules lack peroxidase and acid phosphatase. Some enzymes, e.g. lysozyme, are present in both types of granule.

In mature neutrophils the nucleus is characteristically multilobed with up to six (usually three or four) segments joined by narrow nuclear strands; this is known as the segmented stage. Less mature cells have fewer lobes. The earliest to be released under normal conditions are juveniles (band or stab cells), in which the nucleus is an unsegmented crescent or band. In certain clinical conditions, even earlier stages in neutrophil formation, when cells display indented or rounded nuclei (metamyelocytes or myelocytes) may be released from the bone marrow. In mature cells the edges of the nuclear lobes are often irregular. In females 3% of the nuclei of neutrophils show a conspicuous 'drumstick' formation, which represents the sex chromatin of the inactive X chromosome (Barr body). Neutrophil cytoplasm contains few mitochondria but abundant cytoskeletal elements, including actin filaments, microtubules and their associated proteins, all characteristic of highly motile cells.

Neutrophils are important in the defence of the body against microorganisms. They can phagocytose microbes and small particles in the circulation and, after extravasation, they carry out similar activities in other tissues. They function effectively in relatively anaerobic conditions, relying largely on glycolytic metabolism, and they fulfil an important role in the acute inflammatory phase of tissue injury, responding to chemotaxins released by damaged tissue. Phagocytosis of cellular debris or invading microorganisms is followed by fusion of the phagocytic vacuole with granules, which results in bacterial killing and digestion. Actively phagocytic neutrophils are able to reduce oxygen enzymatically to form reactive oxygen species including superoxide radicals and hydrogen peroxide, which enhance bacterial destruction probably by activation of some of the granule contents (Segal 2005, Nathan 2006). Neutrophils can also produce neutrophil extracellular traps (NETs), which are web-like structures composed of DNA and proteolytic enzymes that can trap bacteria and kill them (Kaplan and Radic 2012).

Phagocytosis is greatly facilitated by circulating antibodies to molecules such as bacterial antigens, which the body has previously encountered. Antibodies coat the antigenic target and bind the plasma complement protein, C1, to their non-variable Fc regions. This activates the complement cascade, which involves some 20 plasma proteins synthesized mainly in the liver, and completes the process of opsonization. The complement cascade involves the sequential cleavage of the complement proteins into a large fragment, which generally binds to the antigenic surface, and a small bioactive fragment, which is released. The final step is the recognition of complement by receptors on the surfaces of neutrophils (and macrophages), which promotes phagocytosis of the organism.

Neutrophils are short-lived; they spend some 6–7 hours circulating in the blood and a few days in connective tissues. The number of circulating neutrophils varies, and often rises during episodes of acute bacterial infection. They die after carrying out their phagocytic role; dead neutrophils, bacteria, tissue debris (including tissue damaged by neutrophil enzymes and toxins) and interstitial fluid form the characteristic, greenish-yellow pus of infected tissue. The colour is derived from the natural colour of neutrophil myeloperoxidase.

Granules may also be released inappropriately from neutrophils. Their enzymes are implicated in various pathological conditions, e.g. rheumatoid arthritis, where tissue destruction and chronic inflammation occur.

Eosinophil granulocytes

Eosinophil granulocytes (eosinophils; for a review, see Rothenberg and Hogan (2006)) are similar in size (12–15 μm), shape and motile

capacity to neutrophils, but are present only in small numbers in normal blood (100–400/ μl). The nucleus has two prominent lobes connected by a thin strand of chromatin. Their cytoplasmic specific granules are uniformly large (0.5 μm) and give the living cell a slightly yellowish colour. The cytoplasm is packed with granules, which are spherical or ellipsoid and membrane-bound. The core of each granule is composed of a lattice of major basic protein, which is responsible for its strong eosinophilic staining properties. The surrounding matrix contains several lysosomal enzymes including acid phosphatase, ribonuclease, phospholipase and a myeloperoxidase unique to eosinophils.

Like other leukocytes, eosinophils are motile. When suitably stimulated, they are able to pass into the extravascular tissues from the circulation. They are typical minor constituents of the dermis, and of the connective tissue components of the bronchial tree, alimentary tract, uterus and vagina. The total lifespan of these cells is a few days, of which some 10 hours is spent in the circulation, and the remainder in the extravascular tissues.

Eosinophil numbers rise (eosinophilia) in worm infestations and also in certain allergic disorders, and it is thought that they evolved as a primary defence against parasitic attack. They have surface receptors for IgE that bind to IgE-antigen complexes, triggering phagocytosis and release of granule contents. However, they are only weakly phagocytic and their most important function is the destruction of parasites too large to phagocytose. This antiparasitic effect is mediated via toxic molecules released from their granules (e.g. eosinophil cationic protein and major basic protein). They also release histaminase, which limits the inflammatory consequences of mast cell degranulation. High local concentrations of eosinophils, e.g. in bronchial asthma and in cutaneous contact sensitivity and allergic eczema, can cause tissue destruction as a consequence of the release of molecules such as collagenase from their granules.

Basophil granulocytes

Slightly smaller than other granulocytes, basophil granulocytes are 10–14 μm in diameter, and form only 0.5–1% of the total leukocyte population of normal blood, with a count of 25–200/ μl . Their distinguishing feature is the presence of large, conspicuous basophilic granules. The nucleus is somewhat irregular or bilobed, and is usually obscured in stained blood smears by the similar colour of the basophilic granules. The granules are membrane-bound vesicles, which display a variety of crystalline, lamellar and granular inclusions: they contain heparin, histamine and several other inflammatory agents, and closely resemble those of tissue mast cells. Both basophils and mast cells have high-affinity membrane receptors for IgE and are therefore coated with IgE antibody. If this binds to its antigen it triggers degranulation of the cells, producing vasodilation, increased vascular permeability, chemotactic stimuli for other granulocytes, and the symptoms of immediate hypersensitivity, e.g. in allergic rhinitis (hay fever). Despite these similarities, basophils and mast cells develop as separate lineages in the myeloid series, from haemopoietic stem cells in the bone marrow. Evidence from experimental animal models suggests that they are closely related (see Fig. 4.12) but studies on mast cell disorders in humans indicate that their lineages diverge from a more distant ancestral progenitor (Kocabas et al 2005). The role of mast cells in the regulation of responses to pain is of interest clinically as a therapeutic target (Chatterjea and Martinov 2015).

Mononuclear leukocytes

Monocytes

Monocytes are the largest of the leukocytes (15–20 μm in diameter) but they form only a small proportion of the total population (2–8% with a count of 100–700/ μl of blood). The nucleus, which is euchromatic, is relatively large and irregular, often with a characteristic indentation on one side. The cytoplasm is pale-staining, particulate and typically vacuolated. Near the nuclear indentation it contains a prominent Golgi complex and vesicles. Monocytes are actively phagocytic cells and contain numerous lysosomes. Phagocytosis is triggered by recognition of opsonized material, as described for neutrophils. Monocytes are highly motile and possess a well-developed cytoskeleton.

Monocytes express class II MHC antigens and share other similarities to tissue macrophages and dendritic cells. Most monocytes are thought to be in transit via the blood stream from the bone marrow to the peripheral tissues, where they give rise to macrophages and dendritic cells; different monocyte subsets may target inflamed tissues. Like other leukocytes, they pass into extravascular sites through the walls of capillaries and venules.

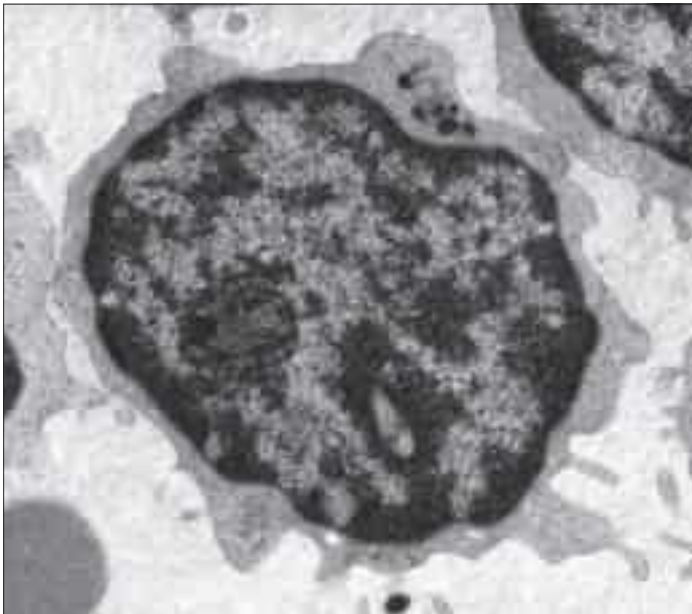


Fig. 4.4 A small, resting lymphocyte in human peripheral blood. The nuclear:cytoplasmic ratio is high and the cytoplasm contains few organelles, indicative of its quiescent state. (Courtesy of Dr Bart Wagner, Histopathology Department, Sheffield Teaching Hospitals, UK.)

Lymphocytes

Lymphocytes (Fig. 4.4; see Figs 4.6, 4.12) are the second most numerous type of leukocyte in adulthood, forming 20–30% of the total population (1500–2700/ μ l of blood). In young children they are the most numerous blood leukocyte. Most circulating lymphocytes are small, 6–8 μ m in diameter; a few are medium-sized and have an increased cytoplasmic volume, often in response to antigenic stimulation. Occasionally, cells up to 16 μ m are seen in peripheral blood. Lymphocytes, like other leukocytes, are found in extravascular tissues (including lymphoid tissue); however, they are the only white blood cells that return to the circulation. The lifespan of lymphocytes ranges from a few days (short-lived) to many years (long-lived). Long-lived lymphocytes are necessary for the maintenance of immunological memory.

Blood lymphocytes are a heterogeneous collection mainly of B and T cells, and consist of different subsets and different stages of activity and maturity. About 85% of all circulating lymphocytes in normal blood are T cells. Primary immunodeficiency diseases can result from molecular defects in T and B lymphocytes (reviewed in Cunningham-Rundles and Ponda (2005)). Included with the lymphocytes, but probably constituting a separate lineage subset, are the natural killer (NK) cells. NK cells most closely resemble large T cells morphologically.

Small lymphocytes (both B and T cells) contain a rounded, densely staining nucleus that is surrounded by a very narrow rim of cytoplasm, barely visible in the light microscope. In the electron microscope (see Fig. 4.4), few cytoplasmic organelles can be seen apart from a small number of mitochondria, single ribosomes, sparse profiles of endoplasmic reticulum and occasional lysosomes; these features indicate a low metabolic rate and a quiescent phenotype. However, these cells become motile when they contact solid surfaces, and can pass between endothelial cells to exit from, or re-enter, the vascular system. They migrate extensively within various tissues, including epithelia (Fig. 4.5).

Larger lymphocytes include T and B cells that are functionally activated or proliferating after stimulation by antigen, and NK cells. They contain a nucleus, which is, at least in part, euchromatic; a basophilic cytoplasm, which may appear granular; and numerous polyribosome clusters, consistent with active protein synthesis. The ultrastructural appearance of these cells varies according to their class and is described below.

B cells

B cells and the plasma cells that develop from them synthesize and secrete antibodies that can specifically recognize and neutralize foreign (non-self) macromolecules (antigens), and can direct various non-lymphocytic cells (e.g. neutrophils, macrophages and dendritic cells) to phagocytose pathogens. B cells differentiate from haemopoietic stem cells in the bone marrow. After deletion of autoreactive cells, the selected B lymphocytes then leave the bone marrow and migrate to peripheral lymphoid sites (e.g. lymph nodes). Here, following stimulation by antigen, they undergo further proliferation and selection, forming

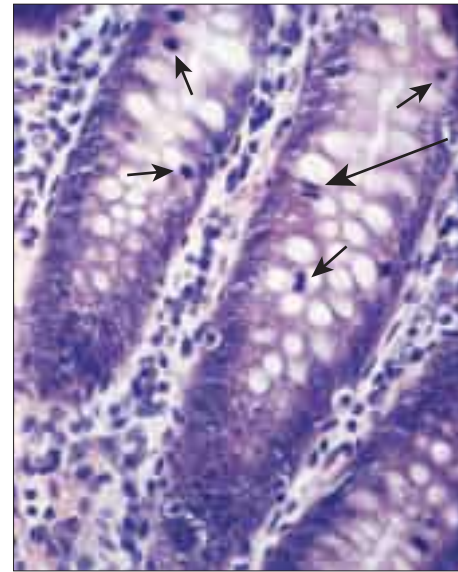


Fig. 4.5 Tubular glands in the appendix, showing intraepithelial lymphocytes (short arrows). A lymphocyte in anaphase is indicated (long arrow).

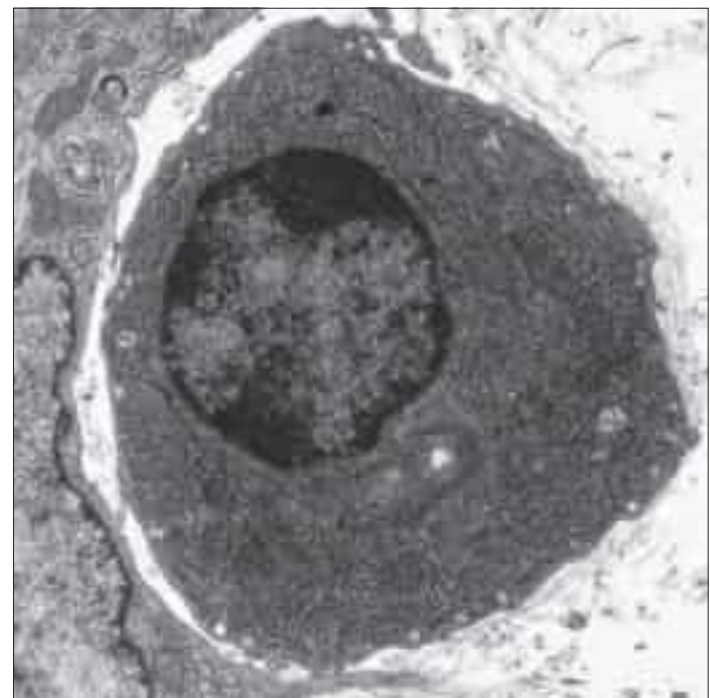


Fig. 4.6 A mature B cell (plasma cell) in human connective tissue. The abundant rough endoplasmic reticulum is typical of a cell actively synthesizing secretory protein, in this case immunoglobulin. The cell to the left is a fibroblast. (Courtesy of Dr Bart Wagner, Histopathology Department, Sheffield Teaching Hospitals, UK.)

germinal centres in the lymphoid tissues. Following this, some B cells differentiate into large basophilic (RNA-rich) plasma cells, either within or outside the lymphoid tissues. Plasma cells produce antibodies in their extensive rough endoplasmic reticulum (Fig. 4.6) and secrete them into the adjacent tissues. They have a prominent pale-staining Golgi complex adjacent to an eccentrically placed nucleus, typically with peripheral blocks of condensed heterochromatin resembling the numerals of a clock (clock-faced nucleus) (see Fig. 4.12). Other germinal centre B cells develop into long-lived memory cells capable of responding to their specific antigens not only with a more rapid and higher antibody output, but also with an increased antibody affinity compared with the primary response.

Antibodies are immunoglobulins, grouped into five classes according to their heavy polypeptide chain. Immunoglobulin G (IgG) forms the bulk of circulating antibodies. Immunoglobulin M (IgM) is normally synthesized early in immune responses. Immunoglobulin A (IgA) is present in breast milk, tears, saliva and other secretions of the alimentary tract, coupled to a secretory piece (a 70 kDa protein) that is synthesized by the epithelial cells. This protects the immunoglobulin from proteolytic degradation and is part of the process by which the molecule is transported across the epithelium; IgA thus contributes to mucosal immunity. IgA deficiency is relatively common, particularly in

some ethnic groups (reviewed in [Woof and Kerr \(2006\)](#)). Immunoglobulin E (IgE) is an antibody which binds to receptors on the surfaces of mast cells, eosinophils and blood basophils; it is found only at low concentrations in the circulation. Immunoglobulin D (IgD) is found together with IgM as a major membrane-bound immunoglobulin on mature, immunocompetent but naïve (prior to antigen exposure) B cells, acting as the cellular receptor for antigen.

When circulating antibodies bind to antigens they form immune complexes. If present in abnormal quantities, these may cause pathological damage to the vascular system and other tissues, either by interfering mechanically with the permeability of the basal lamina (e.g. some types of glomerulonephritis), or by causing local activation of the complement system that generates inflammatory mediators (e.g. C5a), attacks cell membranes and causes vascular disease. In pregnancy, maternal IgG crosses the placenta and confers passive immunity on the fetus. Maternal milk contains secretory immunoglobulins (IgA) that help to combat bacterial and viral organisms in the alimentary tract of the baby during the first few weeks of postnatal life.

T cells

There are a number of subsets of T (thymus-derived) lymphocytes, all progeny of haemopoietic stem cells in the bone marrow. They develop and mature in the thymus, and subsequently populate peripheral secondary lymphoid organs, which they constantly leave and re-enter via the circulation. As recirculating cells, their major function is immune surveillance. Their activation and subsequent proliferation and functional maturation are under the control of antigen-presenting cells. T cells undertake a wide variety of cell-mediated defensive functions that are not directly dependent on antibody activity, and which constitute the basis of cellular immunity. T-cell responses focus on the destruction of cellular targets such as virus-infected cells, certain bacterial infections, fungi, some protozoal infections, neoplastic cells and the cells of grafts from other individuals (allografts, when the tissue antigens of the donor and recipient are not sufficiently similar). Targets may be killed directly by cytotoxic T cells, or indirectly by accessory cells (e.g. macrophages) that have been recruited and activated by cytokine-secreting helper T cells. A third group, regulatory T cells, acts to regulate or limit immune responses.

Functional groups of T cells are classified according to the molecules they express on their surfaces. The majority of cytokine-secreting helper T cells express CD4, while cytotoxic T cells are characterized by CD8. Regulatory T cells co-express CD4 and CD25. The CD (cluster of differentiation) prefix provides a standard nomenclature for all cell-surface molecules. At present, more than 330 different CD antigens have been designated; each one represents a cell surface molecule that can be identified by specific antibodies. Further details of the classification are beyond the scope of this publication and are given in [Male et al \(2012\)](#).

Structurally, T lymphocytes present different appearances depending on their type and state of activity. When resting, they are typically small lymphocytes and are morphologically indistinguishable from B lymphocytes. When stimulated, they become large (up to 15 µm), moderately basophilic cells, with a partially euchromatic nucleus and numerous free ribosomes, rough and smooth endoplasmic reticulum, a Golgi complex and a few mitochondria, in their cytoplasm. Cytotoxic T cells contain dense lysosome-like vacuoles that function in cytotoxic killing.

Cytotoxic T cells

Cytotoxic T lymphocytes (which express CD8) are responsible for the direct cytotoxic killing of target cells (e.g. virus-infected cells); the requirement for direct cell–cell contact ensures the specificity of the response. Recognition of antigen, presented as a peptide fragment on MHC class I molecules, triggers the calcium-dependent release of lytic granules by the T cell. These lysosome-like granules contain perforin (cytolysin), which forms a pore in the target cell membrane. They also contain several different serine protease enzymes (granzymes), which enter the target cell via the perforin pore and induce the programmed cell death (apoptosis; [p. 26](#)) of the target.

Helper T cells

Helper T cells (which express CD4) are characterized by the secretion of cytokines. Two major populations have been identified according to the range of cytokines produced. Th1 helper T cells typically secrete interleukin (IL)-2, tumour necrosis factor alpha (TNF-α) and interferon gamma (IFN-γ), while Th2 cells produce cytokines such as IL-4, IL-5 and IL-13. These two CD4-expressing populations are termed ‘helper’ T cells because one aspect of their function is to stimulate the proliferation and maturation of B lymphocytes and cytotoxic T lymphocytes (mediated via cytokines such as IL-4, IL-2 and IFN-γ), thus enabling

and enhancing the immune responses mediated by those cells. In addition to Th1 and Th2 cells, other subsets of helper T cells have been described. These include Th17 cells (which secrete the cytokine IL-17), implicated in autoimmune diseases ([Stockinger and Veldhoen 2007](#)). Other subsets (Th9, Th22) have been described and it is likely that many more subsets will be characterized in the future ([Jiang and Dong 2013](#)).

Helper T cells are also important in directing the destruction of pathogens by recruiting accessory cells (e.g. macrophages, neutrophils, eosinophils) to the site of infection and by activating their effector functions. This process is tightly coordinated. For example, Th1 helper T cells secrete cytokines that not only attract and activate macrophages but also provide help for B cells and guide their immunoglobulin production to the subclasses that fix complement. Thus these antibodies opsonize the pathogen target, which can then be recognized, ingested and destroyed by the macrophage accessory cells that bear receptors for complement and the Fc region of IgG. These Th1 cells are sometimes referred to as delayed-type hypersensitivity T cells. In contrast, Th2 cells secrete cytokines that induce the development and activation of eosinophils, and also induce B cells to switch their immunoglobulins to non-complement-fixing classes (e.g. IgE). Pathogens such as parasitic worms can then be coated with IgE antibody and hence recognized and destroyed by the effector functions of the eosinophil accessory cells, which bear receptors for the Fc region of IgE.

If helper T-cell activities are non-functional, a state of immunodeficiency results. This means that potentially pathogenic organisms, which are normally kept in check by the immune system, may proliferate and cause overt pathology, e.g. in acquired immune deficiency syndrome (AIDS), where a virus (human immunodeficiency virus, HIV) specifically infects and kills (predominantly) helper T cells, though some antigen-presenting cells are also killed.

Regulatory T cells

A third population of T cells, ‘regulatory’ T or ‘Treg’ cells, are important in controlling the immune response. These CD4⁺, CD25⁺ cells have an immunomodulatory function and can dampen the effector functions of both cytotoxic and helper T cells. Regulatory T cells (natural Tregs) are produced in the thymus and are an important additional mechanism for maintaining self-tolerance ([Safinia et al 2013](#)). Treg function is antigen-specific and depends on direct cell–cell contact. Molecules secreted or induced by Treg cells, such as IL-10 or transforming growth factor beta (TGF-β), also play an important role in mediating Treg suppressive effects on the immune system. Similar induced regulatory T cells can be induced in the periphery and may be important in the induction of oral tolerance to ingested antigens, as well as tolerance to tissue-specific molecules that are not expressed in the thymus ([Schmitt and Williams 2013](#)).

Natural killer (NK) cells

Natural killer (NK) cells have functional similarities to cytotoxic T cells but they lack other typical lymphocyte features and do not express antigen-specific receptors. They normally form only a small percentage of all lymphocyte-like cells and are usually included in the large granular lymphocyte category. When mature, NK cells have a mildly basophilic cytoplasm. Ultrastructurally, their cytoplasm contains ribosomes, rough endoplasmic reticulum and dense, membrane-bound vesicles 200–500 nm in diameter with crystalline cores. These contain the protein perforin, which is capable of inserting holes in the plasma membranes of target cells, and granzymes, which trigger subsequent target cell death by apoptosis. NK cells are activated to kill target cells by a number of factors. They can recognize and kill antibody-coated target cells via a mechanism termed antibody-dependent cell-mediated cytotoxicity (ADCC). They also have receptors that inhibit NK destructive activity when they recognize MHC class I on normal cells. When NK cells detect the loss or downregulation of MHC class I antigens on certain virus-infected cells and some tumour cells, they activate apoptosis-inducing mechanisms that enable them to attack these abnormal cells, albeit relatively non-specifically. For further reading, see [Vivier et al \(2008\)](#) and [Chan et al \(2014\)](#).

PLATELETS

Blood platelets, also known as thrombocytes, are relatively small (2–4 µm across) irregular or oval discs present in large numbers (200,000–400,000/µl) in blood. In freshly harvested blood samples they readily adhere to each other and to all available surfaces, unless the blood is treated with citrate or other substances that reduce the availability of calcium ions. Platelets are anucleate cell fragments, derived from megakaryocytes in the bone marrow. They are surrounded

by a plasma membrane with a thick glycoprotein coat, which is responsible for their adhesive properties. A band of 10 microtubules lies around the perimeter of the platelet beneath the plasma membrane; the microtubules are associated with actin filaments, myosin and other proteins related to cell contraction. The cytoplasm also contains mitochondria, glycogen, scant smooth endoplasmic reticulum, tubular invaginations of the plasma membrane, and three major types of membrane-bound vesicle, designated α , δ and λ granules.

Alpha granules are the largest, and have diameters of up to 500 nm. They contain platelet-derived growth factor (PDGF), fibrinogen and other substances. Delta granules are smaller (up to 300 nm) and contain 5-hydroxytryptamine (5-HT, serotonin) that has been endocytosed from the blood plasma. Lambda granules are the smallest (up to 250 nm) and contain lysosomal enzymes.

Platelets play an important role in haemostasis. When a blood vessel is damaged, platelets become activated, evert their membrane invaginations to form lamellipodia and filopodia, and aggregate at the site of injury, plugging the wound. They adhere to each other (agglutination) and to other tissues. Adhesion is a function of the thick platelet coat and is promoted by the release of adenosine diphosphate (ADP) and calcium ions from the platelets in response to vessel injury. The contents of released α granules, together with factors released from the damaged tissues, initiate a complex sequence of chemical reactions in the blood plasma, which leads to the precipitation of insoluble fibrin filaments in a three-dimensional meshwork, the fibrin clot (see Fig. 4.1). More platelets attach to the clot, inserting extensions of their surfaces, filopodia, deep into the spaces between the fibrin filaments, to which they adhere strongly. The platelets then contract (clot retraction) by actin-myosin interactions within their cytoplasm, and this concentrates the fibrin clot and pulls the walls of the blood vessel together, which limits any further leakage of blood. After repair of the vessel wall, which may be promoted by the mitogenic activity of PDGF, the clot is dissolved by enzymes such as plasmin. Plasmin is formed by plasminogen activators in the plasma, probably assisted by lysosomal enzymes derived from the λ granules of platelets. Platelets typically circulate for 10 days before they are removed, mainly by splenic macrophages.

together to initiate immune responses to foreign antigens (Malhotra et al 2013). These secondary tissues include lymph nodes, spleen, and lymphoid tissue associated with epithelial surfaces (mucosa-associated lymphoid tissue, MALT), e.g. the palatine and nasopharyngeal tonsils, Peyer's patches in the small intestine, lymphoid nodules in the respiratory and urogenital systems, the skin. The microstructure of lymph nodes and of MALT in general is described below. Details of all other lymphoid tissues and organs are included in the descriptions of the appropriate regional anatomy.

Lymphocytes enter secondary lymphoid tissues from the blood, usually by migration through the walls of capillaries or venules (high endothelial venules, HEVs (see Ch. 6)) and leave by the lymphatic system. In the spleen, lymphocyte entry and exit take place via the marginal zone and venous drainage respectively. Antigen-presenting cells (dendritic cells) enter via the lymphatics, bringing with them antigen from peripheral infected sites. In all the secondary tissues there are specific areas where either B or T cells are concentrated. After activation, functionally competent lymphocytes migrate to other sites in the body, where they combat the original infection. Organized lymphoid structures, termed tertiary lymphoid organs, can also develop at sites of chronic inflammation (Stranford and Ruddle 2012).

LYMPH NODES

Lymph nodes are encapsulated centres of antigen presentation and lymphocyte activation, differentiation and proliferation, which are facilitated by complex trafficking of cells and lymphatic flow through the structure (Girard et al 2012). They generate mature, antigen-primed B and T cells, and filter particles, including microbes, from the lymph by the action of numerous phagocytic macrophages. A normal young adult body contains up to 450 lymph nodes, of which 60–70 are found in the head and neck, 100 in the thorax and as many as 250 in the abdomen and pelvis. Lymph nodes are particularly numerous in the neck, mediastinum, posterior abdominal wall, abdominal mesenteries, pelvis and proximal regions of the limbs (axillary and inguinal lymph nodes). By far the greatest number lie close to the viscera, especially in the mesenteries.

Microstructure

Lymph nodes (Fig. 4.7) are small, oval or kidney-shaped bodies, 0.1–2.5 cm long, lying along the course of the lymphatic vessels. Each usually has a slight indentation on one side: the hilum, through which blood vessels enter and leave, and the efferent lymphatic vessel leaves. Several afferent lymphatic vessels enter the capsule around the periphery. Lymph nodes have a highly cellular cortex and a medulla (Fig. 4.8), which contains a network of minute lymphatic channels (sinuses) through which lymph from the afferent lymphatics is filtered, to be

LYMPHOID TISSUES

Lymphocytes are located in many sites in the body, most obviously at strategic sites that are liable to infection, e.g. the oropharynx. The main areas of lymphocyte concentration are classified as primary or secondary lymphoid organs, according to whether they are involved in *de novo* lymphocyte generation (primary lymphoid organs, e.g. bone marrow, thymus) or are the site of mature lymphocyte activation and initiation of an immune response (secondary lymphoid organs, e.g. lymph nodes, spleen).

The secondary or peripheral lymphoid organs are the specialized sites where B and T lymphocytes and antigen-presenting cells come

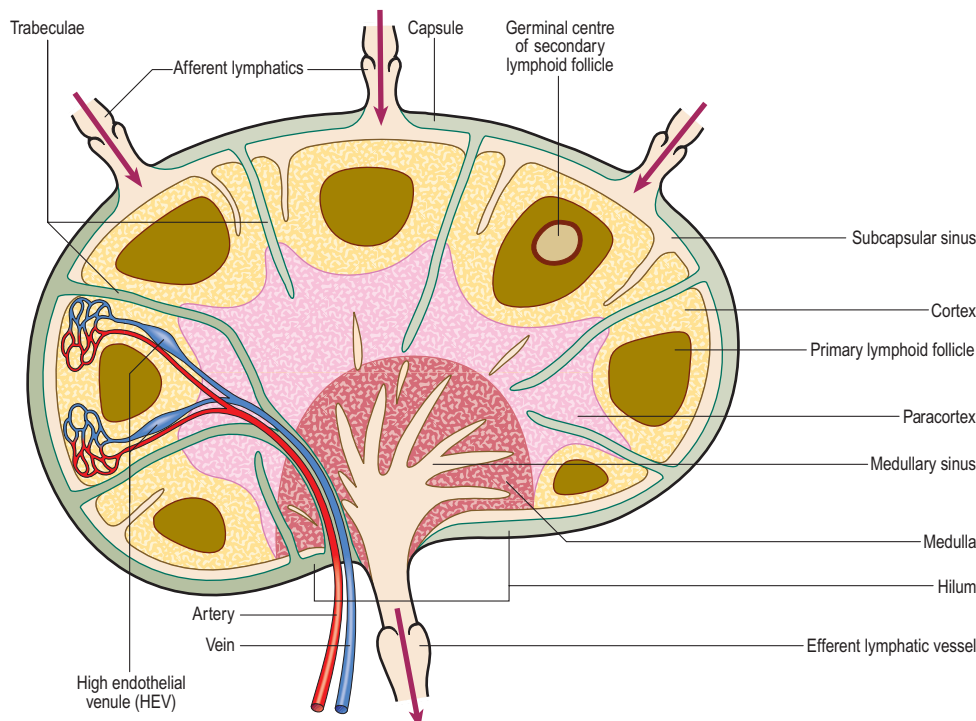


Fig. 4.7 The structure of a lymph node.

All lymphocytes arise from pluripotent haemopoietic stem cells in the bone marrow. The B lymphocyte lineage develops through a series of differentiation stages within the bone marrow. The newly formed B cells then leave through the circulation and migrate to peripheral sites. In contrast, T-lymphocyte development requires the thymus; the bone marrow-derived stem cells must therefore migrate via the blood circulation to the thymus. After their differentiation and maturation into immunocompetent T cells that have survived thymic selection processes (1–3%), they re-enter the circulation and are transported to peripheral sites where they join the pool of naïve lymphocytes that recirculate through the secondary lymphoid organs via blood and lymphatic circulation systems.

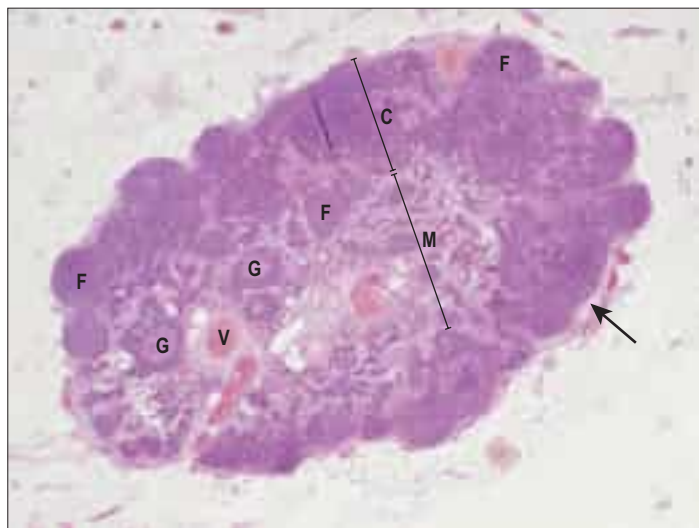


Fig. 4.8 A lymph node (human) showing cortex (C) and medulla (M), and lymphoid follicles (F), some with germinal centres (G). Also shown are the subcapsular sinus (arrow) and medullary blood vessels (V). The dark line (top, centre) is a small crease in the tissue section and is an artefact. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

collected at the hilum by the efferent lymphatic. The cortex is absent at the hilum, where the medulla reaches the surface.

The capsule is composed mainly of collagen fibres, elastin fibres (especially in the deeper layers) and a few fibroblasts. Trabeculae of dense connective tissue extend radially into the interior of the node from the capsule. They are continuous with a network of fine type III collagen (reticulin) fibrils, which branch and interconnect to form a very dense network in the cortex, providing attachment for various cells, mostly dendritic cells, macrophages and lymphocytes. There are fewer fibres in the germinal centres of follicles (see below). Reticulin and the associated proteoglycan matrix are produced by fibroblasts associated with the fibrous network.

Lymphatic and vascular supply

Lymph nodes are permeated by channels through which lymph percolates after its entry from the afferent vessels. The conduit system consists of collagen fibres and associated fibrils surrounded by fibroblast reticular cells, forming a sponge-like reticulum that provides not only spaces for the lymphocytes but also a system for the transport of antigen and signalling molecules (such as chemokines) that control the highly dynamic movement and interaction of the immune cells. Dendritic cells can reach inside the conduits to sample antigen, and then present it to immune cells (Rooszendaal et al 2008).

Afferent lymphatic vessels enter at many points on the periphery, branch to form a dense intracapsular plexus, and then open into the subcapsular sinus, a cavity that is peripheral to the whole cortex except at the hilum (see Fig. 4.7). Numerous radial cortical sinuses lead from the subcapsular sinus to the medulla, where they coalesce as larger medullary sinuses. The latter become confluent at the hilum with the efferent vessel that drains the node. All of these spaces are lined by a continuous endothelium and traversed by fine reticular fibres.

Arteries and veins serving lymph nodes pass through the hilum, giving off straight branches that traverse the medulla and send out minor branches as they do so. In the cortex, arteries form dense arcades of arterioles and capillaries in numerous anastomosing loops, eventually returning to highly branched venules and veins. Capillaries are especially profuse around the follicles, which contain fewer vessels. Postcapillary HEVs are abundant in the paracortical zones. They form an important site of blood-borne lymphocyte extravasation into lymphoid tissue, apparently by migration through labile endothelial tight junctions (occluding junctions, zonulae adherentes). The density of the capillary beds increases greatly when lymphocytes multiply in response to antigenic stimulation. Veins leave a node through its principal trabeculae and capsule, and drain them and the surrounding connective tissue.

Cells and cellular zones of lymph nodes

Although most of the cells in a lymph node are B and T lymphocytes, their distribution is not homogeneous. In the cortex, cells are densely

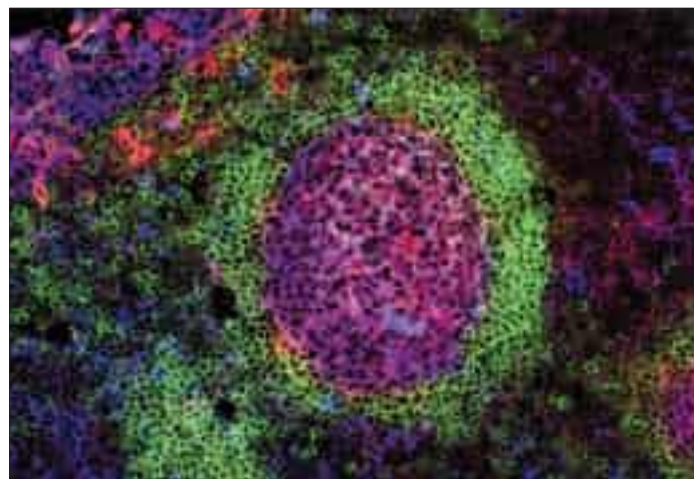


Fig. 4.9 A germinal centre in a lymphoid follicle of the human palatine tonsil, immunolabelled to show CD38-positive B cells in the germinal centre (red), IgD-positive naïve B cells (green) in the mantle zone and activated, transferrin receptor (CD71)-positive cells of various lineages (blue). (Courtesy of Dr Cécile Chalouni, Ludwig Institute for Cancer Research, Yale University School of Medicine, USA.)

packed and in the outer cortical area they form lymphoid follicles or nodules (see Fig. 4.8), which are populated mainly by B cells and specialized follicular dendritic cells (FDCs) (see Fig. 4.15). The number, degree of isolation and staining characteristics of follicles vary according to their state of antigenic stimulation. A primary follicle is uniformly populated by small, quiescent lymphocytes, whereas a secondary follicle has a germinal centre (Fig. 4.9), composed mainly of antigen-stimulated B cells, which are larger, less deeply staining and more rapidly dividing than those at its periphery.

The role of the germinal centre is to provide a microenvironment that allows the affinity maturation of the B-cell response, so that as the immune response progresses, the affinity or strength with which antibodies bind their antigen increases (Shlomchik and Weisel 2012, Victora and Nussenzweig 2012). There are several zones in the germinal centre where this is allowed to happen. In the 'dark zone', the B cells (centroblasts) undergo rapid proliferation, which is associated with hypermutation of their antibody molecules. They then move into the 'light zone' (as centrocytes), where they can interact with the FDCs, which carry intact unprocessed antigen on their surface in the form of immune complexes (Rezk et al 2013). The centrocytes compete for binding to the antigen; those whose antibody has the highest affinity survive and the rest die. T cells are also present, helping the survival of the B cells and inducing class switching. Macrophages in the germinal centre phagocytose apoptotic lymphocytes (e.g. those B cells that die as part of the process of affinity maturation), and consequently macrophage cytoplasm becomes filled with engulfed lipid and nuclear debris forming sparkling intracellular inclusions (leading to the term tingible body macrophage).

The mantle zone (see Fig. 4.9) is produced as surrounding cells are marginalized by the rapidly growing germinal centre. It is populated by cells similar to those found in primary follicles: mainly quiescent B cells with condensed heterochromatic nuclei and little cytoplasm (hence the deeply basophilic staining of this region in routine preparations; Fig. 4.10), a few helper T cells, FDCs and macrophages. After numerous mitotic divisions the selected B cells give rise to small lymphocytes, some of which become memory B cells and leave the lymph node to join the recirculating pool, while others leave to mature as antibody-secreting plasma cells either in the lymph node medulla or in peripheral tissues.

The deep cortex or paracortex lies between the cortical follicles and the medulla, and is populated mainly by T cells, which are not organized into follicles. Both CD4 and CD8 T-cell subsets are present. The paracortex also contains interdigitating dendritic cells. These dendritic cells include Langerhans cells from the skin and other squamous epithelia, which have migrated as veiled cells via the afferent lymphatics into the draining lymph nodes (see Fig. 4.14). Their role is to present processed antigen to T cells. The region expands greatly in T cell-dominated immune responses, when its cells are stimulated to proliferate and disperse to peripheral sites.

In the medulla, lymphocytes are much less densely packed in irregular, branching medullary cords between which the reticulin network is easily seen. Other cells include macrophages, which are more numer-



Fig. 4.10 A germinal centre in a follicle of mucosa-associated lymphoid tissue (MALT) in the mucosa and submucosa of the appendix. The bases of tubular glands of the mucosal epithelium are seen in the upper field.

ous in the medulla than in the cortex, plasma cells and a few granulocytes.

MUCOSA-ASSOCIATED LYMPHOID TISSUE (MALT)

Large amounts of unencapsulated lymphoid tissue exist in the walls of the gastrointestinal, respiratory, reproductive and urinary tracts, and in the skin; they are collectively termed mucosa-associated lymphoid tissue (MALT), most of which is found in the gut as gut-associated lymphoid tissue (GALT) (Koboziev et al 2010).

Throughout the body, MALT includes an extremely large population of lymphocytes, reflecting the size of the gastrointestinal tract. Lymphoid cells are located in the lamina propria and in the submucosa as discrete follicles or nodules. More scattered cells, derived from these follicles, are found throughout the lamina propria and in the base of the epithelium (see Figs 4.5, 4.10). MALT includes macroscopically visible lymphoid masses, notably the peripharyngeal lymphoid ring of tonsillar tissue (palatine, nasopharyngeal, tubal and lingual), and the Peyer's patches of the small intestine, all of which are described elsewhere. Most MALT consists of microscopic aggregates of lymphoid tissue, which lack a fibrous capsule. Lymphocyte populations are supported mechanically by a network of fine type III collagen (reticulin) fibres and associated fibroblasts, as they are in lymph nodes.

In common with lymph nodes, MALT provides centres for the activation and proliferation of B and T lymphocytes in its follicles and para-follicular zones, respectively. The function of cells in these zones, including antigen-presenting cells (follicular dendritic cells and interdigitating dendritic cells) and macrophages, as well as T and B cells, is similar to that found in lymph nodes. The close proximity of lymphocytes within MALT to an epithelial surface facilitates their access to pathogens. MALT lacks afferent lymphatic vessels. Lymphocytes migrate into MALT through its HEVs and leave mainly via its efferent lymphatics, which drain interstitial fluid as lymph; the lymphocyte population in MALT is not fixed. Migration from MALT follows a different route from the major peripheral route of recirculation. After antigen activation, lymphocytes travel via the regional lymph nodes to disperse widely along mucosal surfaces to provide protective T- and B-cell immunity.

Follicle-associated epithelium

The epithelium covering MALT varies in type according to its location. It is unusual in possessing cells that are involved in sampling antigens and transferring them to antigen-presenting cells in the underlying tissues; appropriate clones of T and B cells in local lymphoid tissues are then activated and amplified prior to their exit via the lymphatics. Specialized epithelial cells in the small and large intestine have characteristic short microvilli on their luminal surfaces and are known as microfold (M) cells (Kraehenbuhl and Neutra 2000); in the palatine

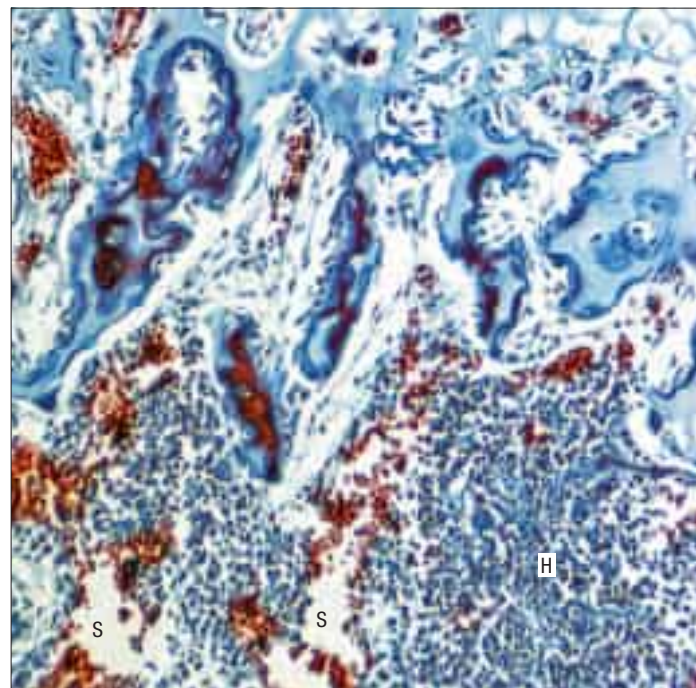


Fig. 4.11 Haemopoietic tissue (H) in the marrow cavity of a fetal long bone undergoing endochondral ossification (top). Islands of densely packed nucleated haemopoietic cells of different lineages are separated by large vascular sinusoids (S), which are filled with mature red blood cells in the general circulation.

tonsils they include modified stratified squamous reticulated epithelial cells (see p. 576). The main function of B lymphocytes in MALT is to produce IgA for secretion into the lumen of the tracts that they line (Cerutti et al 2011).

Many of the lymphocytes migrating between cells in the basal regions of epithelia (see Fig. 4.5) are effector cytotoxic and helper T cells that have already been selected in lymphoid nodules and are engaged in immune responses. Similar cells, and activated IgA-producing B cells and plasma cells, are also scattered throughout the entire mucosal lamina propria.

HAEMOPOIESIS

Postnatally, blood cells are formed primarily in the bone marrow. Other tissues, particularly the spleen and liver, may develop haemopoietic activity once more, if production from the marrow is inadequate.

BONE MARROW

Bone marrow is a soft pulpy tissue that is found in the marrow cavities of all bones (Fig. 4.11) and even in the larger Haversian canals of lamellar bone. It differs in composition in different bones and at different ages, and occurs in two forms: yellow and red marrow. In old age the marrow of the cranial bones undergoes degeneration and is then termed gelatinous marrow.

Yellow marrow

Yellow marrow consists of a framework of connective tissue that supports numerous blood vessels and cells, most of which are adipocytes. A small population of typical red marrow cells persists and may be reactivated when the demand for blood cells becomes sufficiently great.

Red marrow

Red marrow is found throughout the skeleton in the fetus and during the first years of life. After about the fifth year the red marrow, which represents actively haemopoietic tissue, is gradually replaced in the long bones by yellow marrow. The replacement starts earlier, and is generally more advanced, in the more distal bones. By 20–25 years of age, red marrow persists only in the vertebrae, sternum, ribs, clavicles, scapulae, pelvis and cranial bones, and in the proximal ends of the femur and humerus.

Red bone marrow consists of a network of loose connective tissue, the stroma, which supports clusters of haemopoietic cells (haemopoietic cords or islands) and a rich vascular supply in which large, thin-walled sinusoids are the main feature (see Fig. 4.11). The vascular supply is derived from the nutrient artery to the bone, which ramifies in the bone marrow, and terminates in thin-walled arterioles from which the sinusoids arise. These, in turn, drain into disproportionately large veins. Lymphatic vessels are absent from bone marrow. The stroma contains a variable amount of fat, depending on age, site and the haematological status of the body, and small patches of lymphoid tissue are also present. Marrow thus consists of vascular and extravascular compartments, both enclosed within a bony framework from which they are separated by a thin layer of endosteal cells.

Stroma

Stroma is composed of a delicate network of fine type III collagen (reticulin) fibres secreted by highly branched, specialized fibroblast-like cells (reticular cells) derived from embryonic mesenchyme. When haemopoiesis stops, as occurs in most limb bones in adult life, these cells (or closely related cells) become distended with lipid droplets and fill the marrow with yellow fatty tissue (yellow marrow). If there is a later demand for haemopoiesis, the stellate stromal cells reappear. The stroma also contains numerous macrophages attached to extracellular matrix fibres. These cells actively phagocytose cellular debris created by haemopoietic development, especially the extruded nuclei of erythroblasts, remnants of megakaryocytes and cells that have failed the B-lymphocyte selection process. Stromal cells play a major role in the control of haemopoietic cell differentiation, proliferation and maturation.

Marrow sinusoids are lined by a single layer of endothelial cells, supported by reticulin on their basal surfaces. Although the endothelial cells are interconnected by tight junctions, their cytoplasm is extremely thin in places, and the underlying basal lamina is discontinuous. The passage of newly formed blood cells from the haemopoietic compartment into the blood stream appears to occur through an interactive process with the endothelium, producing temporary apertures (large fenestrae) in their attenuated cytoplasm.

Haemopoietic tissue

Cords and islands of haematogenous cells consist of clusters of immature blood cells in various stages of development; several different cell lineages are typically represented in each focal group. One or more macrophages lie at the core of each such group of cells. These macrophages engage in phagocytic functions, are important in transferring iron to developing erythroblasts for haemoglobin synthesis, and may play a role, with other stromal cells, in regulating the rate of cell proliferation and maturation of the neighbouring haemopoietic cells.

CELL LINEAGES

Haemopoietic stem cells

Within the adult marrow there is a very small number (0.05% of haemopoietic cells) of self-renewing, pluripotent stem cells that are capable of giving rise to all blood cell types, including lymphocytes (Fig. 4.12). Although they cannot be identified morphologically in the marrow, they can be recognized in aspirates by the expression of specific cell-surface marker proteins (e.g. CD34). It is thought that haemopoietic stem cells occupy specific environmental niches in the marrow associated with the endosteum of trabecular bone or with sinusoidal endothelium, and that their microenvironment is important in homeostasis, the balance between self-renewal and differentiation. Stem cells can also be found (at lower concentrations) in the peripheral blood, particularly after treatment with appropriate cytokines.

Progressively more lineage-restricted committed progenitor cells develop from these ancestors (see Laiosa et al (2006) for a review) to produce the various cell types found in peripheral blood. The committed progenitor cells are often termed colony-forming units (CFU) of the lineage(s), e.g. CFU-GM cells give rise, after proliferation, to neutrophil granulocytes, monocytes and certain dendritic cells, whereas CFU-E produce only erythrocytes. Each cell type undergoes a period of maturation in the marrow, often accompanied by several structural changes, before release into the general circulation. In some lineages, e.g. the erythroid series, the final stages of maturation take place in the circulation, whereas in the monocytic lineage, they occur after the cells have left the circulation and entered peripheral tissues where they differentiate into macrophages and some dendritic cells.

To generate a complete set of blood cells from a single pluripotent cell may take some months. The later progenitor cells form mature cells of their particular lineages more quickly. However, because they are not self-renewing, grafts of these later cells eventually fail because the cells they produce all ultimately die. This is of considerable importance in bone marrow replacement therapy. The presence of pluripotent stem cells in the donor marrow is essential for success; only 5% of the normal number are needed to repopulate the marrow. Following replacement therapy, T lymphocytes reconstitute more slowly than the other haemopoietic lineages, reflecting the progressive reduction in size of the thymus with age (chronic involution).

Within the bone marrow there is also a population of mesenchymal stem cells that can differentiate into a wide variety of non-haemopoietic cells. These pluripotent stem cells can also be found in the circulation, and are being investigated for their use in repairing damaged organs (see Commentary 1.2).

Lymphocytes

Lymphocytes are a heterogeneous group of cells that may share a common ancestral lymphoid progenitor cell, distinct from the myeloid progenitor cell (see Fig. 4.12). The first identifiable progenitor cell is the lymphoblast, which divides several times to form prolymphocytes; both cells are characterized by a high nuclear:cytoplasmic ratio. B cells undergo differentiation to their specific lineage subset entirely within the bone marrow and migrate to peripheral or secondary lymphoid tissues as naïve B cells, ready to respond to antigen. However, T cells require the specialized thymic microenvironment for their development. During fetal and early postnatal life, and subsequently at lower levels throughout life, progenitor cells migrate to the thymus where they undergo a process of differentiation and selection as T cells, before leaving to populate secondary lymphoid tissues.

B-cell development

B cells start their development in the subosteal region of the bone marrow and move centripetally as differentiation progresses. Their development entails the rearrangement of immunoglobulin genes to create a unique receptor for antigen on each B cell, and the progressive expression of cell-surface and intracellular molecules required for mature B-lymphocyte function. Autoreactive cells that meet their self-antigen within the bone marrow are eliminated. Overall, some 25% of B cells successfully complete these developmental and selection processes; those that fail die by apoptosis and are removed by macrophages. Bone marrow stromal cells (fibroblasts, fat cells and macrophages) express cell-surface molecules and secrete cytokines that control B-lymphocyte development. The mature naïve B lymphocytes leave via the central sinuses. They express antigen receptors (immunoglobulin) of IgM and IgD classes. Class switching to IgG, IgA and IgE occurs in the periphery following antigen activation in response to signals from T helper cells.

T-cell (thymocyte) development

T cells develop within the thymus from blood-borne, bone marrow-derived progenitors that enter the thymus via HEVs at the corticomedullary junction. They first migrate to the outer (subcapsular) region of the thymic cortex and then, as in the bone marrow, move progressively inwards towards the medulla as development continues. T-cell development involves gene rearrangements in the T-cell receptor (TcR) loci to create unique receptors for antigen on each cell, together with the progressive expression of molecules required for mature T-cell function. Selection of the receptor repertoire is more stringent for T cells than for B cells because of the way in which mature T cells recognize peptides derived from protein antigens presented in conjunction with specific molecules of the major histocompatibility complex (MHC) expressed on the surfaces of cells. Thus mature CD8 (cytotoxic) T cells recognize antigen in the form of short peptides complexed with the polymorphic MHC class I molecules, while CD4 (helper/regulatory) T cells recognize the peptides in the context of MHC class II molecules. As the TcR recognizes both the peptide and the MHC molecule, the T cell will only recognize peptides bound to their own (self) type of MHC; it will not 'see' peptides in combination with allelically different MHC molecules (i.e. those from other individuals). This is termed MHC restriction of T-cell recognition. Selection of T cells in the thymus must ensure the survival of those T cells that can respond only to foreign antigens, bound to their own (self) class of MHC molecule. Cells that are incapable of binding to self MHC molecules, or which bind to self-antigens, are eliminated by apoptotic cell death (see p. 26); it is estimated that up to 95% of T-cell progenitors undergo apoptosis in this way. Cells that express an appropriate TcR and have effective MHC-restricted

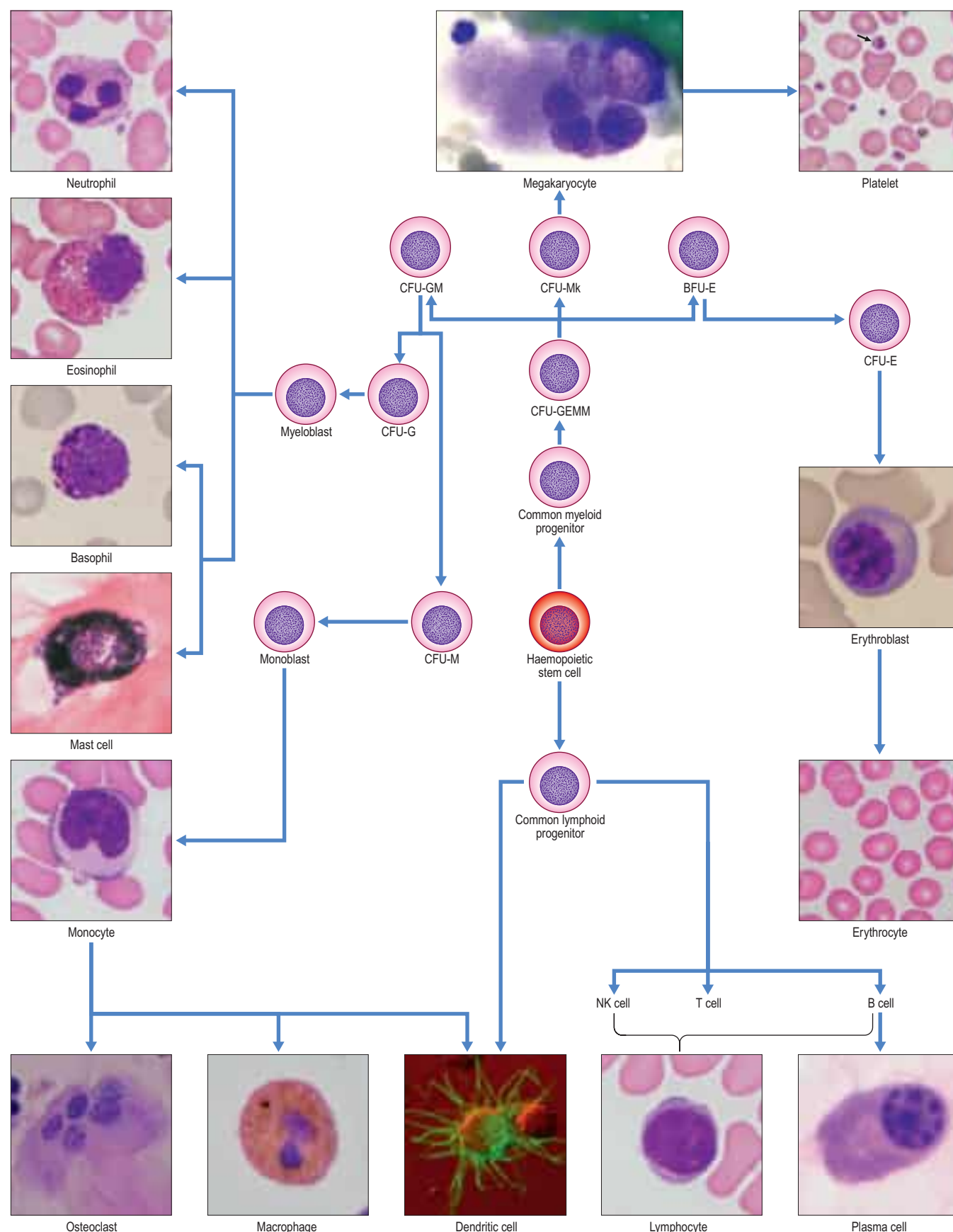


Fig. 4.12 The origins and lineage relationships of haemopoietically derived cells of the immune system. Mature cells and selected progenitors (all human) are illustrated (magnifications vary). The dendritic cell was cultured from peripheral blood, immunolabelled to show HLA-DR and photographed using Nomarski optics. The megakaryocyte and erythroblast are from a bone marrow smear, stained with May-Grünwald-Giemsa (MGG); the remaining cells illustrated are from peripheral blood smears (Wright's stain), sections of connective tissue (plasma cell, mast cell), bone (osteoclast) and lung alveolus (macrophage). Platelets (one is arrowed) are subcellular fragments of bone marrow megakaryocytes. Note that circulating small lymphocytes cannot be classified further with routine staining methods. For further explanation of cellular structure and staining properties, along with abbreviations, see the text. (Dendritic cell image courtesy of Dr Cécile Chalouni, Ludwig Institute for Cancer Research, Yale University School of Medicine, USA. All other images courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

binding properties survive to become mature, naïve T cells that leave the thymus and populate the periphery.

Thymic stromal cells play a crucial role in T-cell development and selection. Thymic epithelial cells in the cortex express both MHC class I and II molecules and are unique in their ability to select T cells that recognize self MHC with a moderate affinity (positive selection). Deletion of self-antigen reactive cells (negative selection) is mainly controlled by thymic dendritic cells located at the corticomedullary junction and in the medulla, although the epithelium can also perform this function. Apoptotic thymocytes are removed by thymic macrophages. The role of the thymic epithelium in thymocyte differentiation is complex and involves cell–cell contact as well as the secretion of soluble mediators such as cytokines, chemokines, neuroactive peptides (e.g. somatostatin) and thymic hormones (e.g. thymulin). Thymic fibroblasts and the extracellular matrix also play a role.

Erythrocytes

Erythrocytes and granulocytes belong to the myeloid lineage. The earliest identifiable erythroid progenitor cells are capable of rapid bursts of cell division to form numerous daughter cells; they have thus been named burst-forming units of the erythroid line (BFU-E; see Fig. 4.12). They give rise to the CFU-E, which, with their immediate progeny, are sensitive to the hormone erythropoietin. This hormone, produced in the kidney, induces further differentiation along the erythroid line.

The first readily identifiable cell of the erythroid series is the proerythroblast, which is a large (about 20 µm) cell with a large euchromatic nucleus and a moderately basophilic cytoplasm. It also responds to erythropoietin. The proerythroblast contains small amounts of ferritin and bears some of the protein spectrin on its plasma membrane. Proerythroblasts proliferate to produce smaller (12–16 µm) basophilic erythroblasts, rich in ribosomes, in which haemoglobin-RNA synthesis begins. The cytoplasm becomes partially, and then uniformly, eosinophilic (the polychromatic erythroblast and orthochromatic erythroblast respectively). These cells are only 8–10 µm in diameter and contain very little cytoplasmic RNA. The nucleus becomes pyknotic (dense, deep-staining, shrunken) and is finally extruded from the cell, leaving an anucleate reticulocyte, which enters a sinusoid. Its reticular staining pattern, visible using special stains, results from residual cytoplasmic RNA, which is usually lost within 24 hours of entering the peripheral blood circulation. Reticulocyte numbers in peripheral blood are therefore a good indicator of the rate of red-cell production. The whole process of erythropoiesis takes 5–9 days.

Granulocytes

Granulocyte formation involves major changes in nuclear morphology and cytoplasmic contents, which are best known for the neutrophil. Initially, myeloid progenitor cells transform into large (10–20 µm) myeloblasts that are similar in general size and appearance to proerythroblasts. These proliferative cells have large euchromatic nuclei and lack cytoplasmic granules. They differentiate into slightly larger promyelocytes, in which the first group of specific proteins is synthesized in the rough endoplasmic reticulum and Golgi apparatus. The proteins are stored in large (0.3 µm) primary (non-specific) granules, which are large lysosomes containing acid phosphatase. Smaller secondary (specific) granules are formed in the smaller myelocyte, which is the last proliferative stage. The nucleus is typically flattened or slightly indented on one side in myelocytes.

In the next, metamyelocyte, stage, the cell size (10–15 µm) decreases, the nucleus becomes heterochromatic and horseshoe-shaped, and protein synthesis almost stops. As the neutrophil is released, the nucleus becomes first heavily indented (the juvenile stab or band form), and subsequently segmented into up to six lobes, characteristic of the mature neutrophil. The whole process usually takes 7 days to complete (3 days proliferating and 4 days maturing). Neutrophils may then be stored in the marrow for a further 4 days, depending on demand, before their final release into the circulation.

Eosinophils and basophils pass through a similar sequence but their nuclei do not become as irregular as that of the neutrophil. It is thought that these cells each arise from distinct colony-forming units, which are separate from the CFU-GM.

Monocytes

Monocytes are formed in the bone marrow. Monocytes and neutrophils appear to be closely related cells; together with some of the antigen-presenting dendritic cells, they arise from a shared progenitor, the

colony-forming unit for granulocytes and macrophages (CFU-GM). Different colony-stimulating factors (CSF) act on the common progenitor to direct its subsequent differentiation pathway. Monocyte progenitors pass through a proliferative monoblast stage (14 µm) and then form differentiating promonocytes, which are slightly smaller cells in which production of small lysosomes begins. After further divisions, monocytes (up to 20 µm) are released into the general circulation. Most migrate into perivascular and extravascular sites, which they then populate as macrophages, while others may give rise to certain dendritic cells, including Langerhans cells.

Platelets

Platelets arise in a unique manner by the shedding of thousands of cytoplasmic fragments from the tips of processes of megakaryocytes in the bone marrow. The first detectable cell of this line is the highly basophilic megakaryoblast (15–50 µm), followed by a promegakaryocyte stage (20–80 µm), in which synthesis of granules begins. Finally, the fully differentiated megakaryocyte, a giant cell (35–160 µm) with a large, dense, polyploid, multilobed nucleus, appears. Once differentiation is initiated from the CFU-Meg, DNA replicates without cytoplasmic division (endoreduplication), and the chromosomes are retained within a single polyploid nucleus that may contain up to 256n chromosomes (where n is the haploid complement present in gametes). Megakaryocyte lineage characteristics and disorders are reviewed in [Sun et al \(2006\)](#).

The cytoplasm contains fine basophilic granules and becomes partitioned into proplatelets by invaginations of the plasma membrane. These are seen ultrastructurally as a network of tubular profiles, which coalesce to form cytoplasmic islands 3–4 µm in diameter. Individual platelets are shed into the circulation from a long, narrow process of megakaryocyte cytoplasm that is protruded through an aperture in the sinusoidal endothelium.

PHAGOCYTES AND ANTIGEN-PRESENTING CELLS

Macrophages and neutrophils (see above) are specialized phagocytes. Certain dendritic cells (see Fig. 4.12), e.g. Langerhans cells of the skin and other stratified squamous epithelia, are 'professional' antigen-presenting cells (APCs); they take up foreign material by endocytosis and macropinocytosis, and are uniquely capable of efficiently activating naïve as well as mature T lymphocytes. Macrophages are also able to process and present antigen to lymphocytes, but are less effective than dendritic cells. In addition, they play an important role in the effector arm of the immune response, clearing the infectious agent by phagocytosis. The third major cell type involved in antigen presentation and T-cell activation is the B lymphocyte, which is particularly efficient at taking up antigen that binds to its surface immunoglobulin (see above). Follicular dendritic cells of lymph nodes, MALT and the spleen are capable of presenting non-processed antigen to B lymphocytes, but are not classic APCs because they cannot present antigen to helper T cells.

APCs endocytose antigen, digest it intracellularly, mostly to peptide fragments, and present the fragments on their surfaces, generally in conjunction with MHC class II molecules. (Class II molecules are normally found only on APCs, although many other cells can express class II molecules in inflammatory situations.) Recognition of foreign antigen is controlled by a variety of APC cell-surface receptors: Fc and complement receptors mediate uptake of opsonized material, while pattern recognition receptors of the innate immune system, e.g. Toll-like receptors and scavenger receptors, directly recognize pathogen-derived molecules.

MACROPHAGES

The mononuclear phagocyte system consists of the blood monocytes, from which the other types are derived, and various tissue macrophages, some of which have tissue-specific names. Certain dendritic cells are sometimes included in the mononuclear phagocyte system; although they share a common lineage ancestor, they appear to form a discrete branch of the family tree. Most monocytes and macrophages express class II MHC molecules.

Macrophages are very variable in size (generally 15–25 µm) and are found in many tissues of the body, where they constitute a heterogeneous family of cells (reviewed in [Gordon and Taylor \(2005\)](#)). They are migrant cells in all general connective tissues, in bone marrow and all lymphoid tissues, and include alveolar macrophages in the lung and Kupffer cells in liver sinusoids. Macrophages often aggregate in

subserous connective tissue of the pleura and peritoneum, where they are visible as milky spots near small lymphatic trunks. They cluster around the terminations of small (penicillar) arterioles in the spleen and are distributed, more diffusely, throughout the splenic cords.

Osteoclasts in bone are closely related to macrophages; they are syncytial cells derived from the fusion of up to 30 progenitor monocytes in bone tissue, where they differentiate further. Microglia of the central nervous system originate from an embryonic monocyte precursor and migrate into the central nervous system during its development. They differ from macrophages in that normally they are quiescent cells in which MHC class II expression is downregulated, and they display little phagocytic activity.

Macrophages vary in structure depending on their location in the body. All have a moderately basophilic cytoplasm containing some rough and smooth endoplasmic reticulum, an active Golgi complex and a large, euchromatic and somewhat irregular nucleus. These features are consistent with an active metabolism: synthesis of lysosomal enzymes continues in mature cells. All macrophages have irregular surfaces with protruding filopodia and they contain varying numbers of endocytic vesicles, larger vacuoles and lysosomes. Some macrophages are highly motile, whereas others tend to remain attached and sedentary, e.g. in hepatic and lymphoid sinuses. Within connective tissues, macrophages may fuse to form large syncytia (giant cells) around particles that are too large to be phagocytosed, or when stimulated by the presence of infectious organisms, e.g. *Mycobacterium tuberculosis*.

When blood-borne monocytes enter the tissues through the endothelial walls of capillaries and venules, they can undergo a limited number of rounds of mitosis as tissue macrophages before they die and are replaced from the bone marrow, typically after several weeks. There is some evidence that alveolar macrophages of the lung are able to undergo many more mitoses than other macrophages.

Phagocytosis

The uptake of particulate material and microorganisms is carried out by macrophages in many tissues and organs. When present in general connective tissue, they ingest and kill invading microorganisms and remove debris that has been produced as a consequence of tissue damage. They recognize, engulf and rapidly ingest apoptotic cells in all situations; the mechanism of apoptotic cell uptake does not activate the phagocyte for antigen presentation, and so the process is immunologically silent. In the lung, alveolar macrophages constantly patrol the respiratory surfaces, to which they migrate from pulmonary connective tissue (Fig. 4.13). They engulf inhaled particles, including bacteria, surfactant and debris, and many enter the sputum (hence their alternative names, dust cells or, in cardiac disease, heart failure cells, which are full of extravasated erythrocytes). They perform similar scavenger functions in the pleural and peritoneal cavities. In lymph nodes, macrophages line the walls of sinuses and remove particulate matter from lymph as it percolates through them. In the spleen and liver, macrophages are involved in particle removal and in the detection and destruction of aged or damaged erythrocytes. They begin the degradation of haemoglobin for recycling iron and amino acids.

Macrophages bear surface receptors for the Fc portions of antibodies and for the fragments of the C3 component of complement. Phagocytic

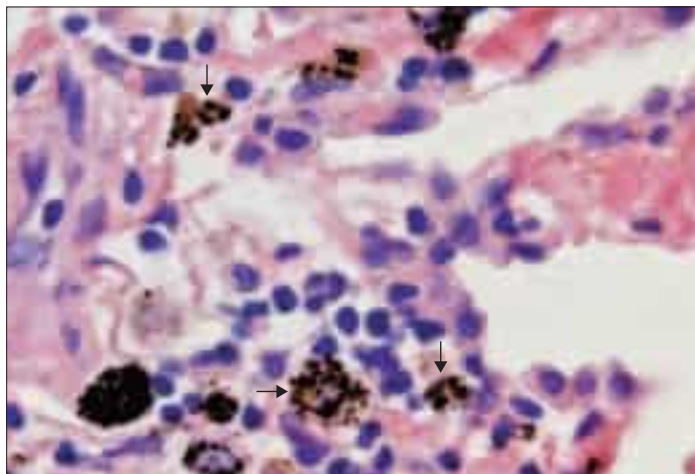


Fig. 4.13 Alveolar macrophages (dust cells, arrows) containing ingested carbon particles, in alveoli and interalveolar septa of the human lung. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

activity is greatly increased when the target has been coated in antibody (opsonized) or complement, or both. Once phagocytosis has occurred, the vacuole bearing the ingested particle fuses with endosomal vesicles, which contain a wide range of lysosomal enzymes, including many hydrolases, and oxidative systems capable of rapid bacteriocidal action. These activities are much enhanced when macrophages are stimulated (activated macrophages) by cytokines, e.g. IFN- γ , which are secreted by other cells of the immune system, especially T lymphocytes.

Close antibody-mediated binding may initiate the release of lysosomal enzymes on to the surfaces of the cellular targets to which the macrophages bind. This process of cytotoxicity is also used by other cells, including neutrophils and eosinophils, particularly if the targets are too large to be phagocytosed (e.g. nematode worm parasites).

Secretory activities

Activated macrophages can synthesize and secrete various bioactive substances, e.g. IL-1, which stimulate the proliferation and maturation of other lymphocytes, greatly amplifying the reaction of the immune system to foreign antigens. They also synthesize TNF- α , which is able to kill small numbers of neoplastic cells. TNF- α depresses the anabolic activities of many cells in the body, and may be a major factor mediating cachexia (wasting), which typically accompanies more advanced cancers. Other macrophage products include plasminogen activator, which promotes clot removal; various lysosomal enzymes; several complement and clotting factors; and lysozyme (an antibacterial protein). In pathogenesis, these substances may be released inappropriately and damage healthy tissues, e.g. in rheumatoid arthritis and various other inflammatory conditions.

DENDRITIC CELLS

There are two distinct groups of dendritic cell: myeloid dendritic cells (also known as classic dendritic cells) and plasmacytoid dendritic cells (Liu 2001, Merad et al 2013, Gerlach et al 2013). These cells can arise from both common lymphoid progenitors and common myeloid progenitors. Both groups of cells are involved in antigen presentation, though have somewhat different functional roles in controlling both the adaptive and innate immune systems. The myeloid dendritic cells are professional APCs, which are able to process and present antigen to T lymphocytes, including naïve T cells. They are present as immature dendritic cells in the epidermis of the skin (Fig. 4.14) and other

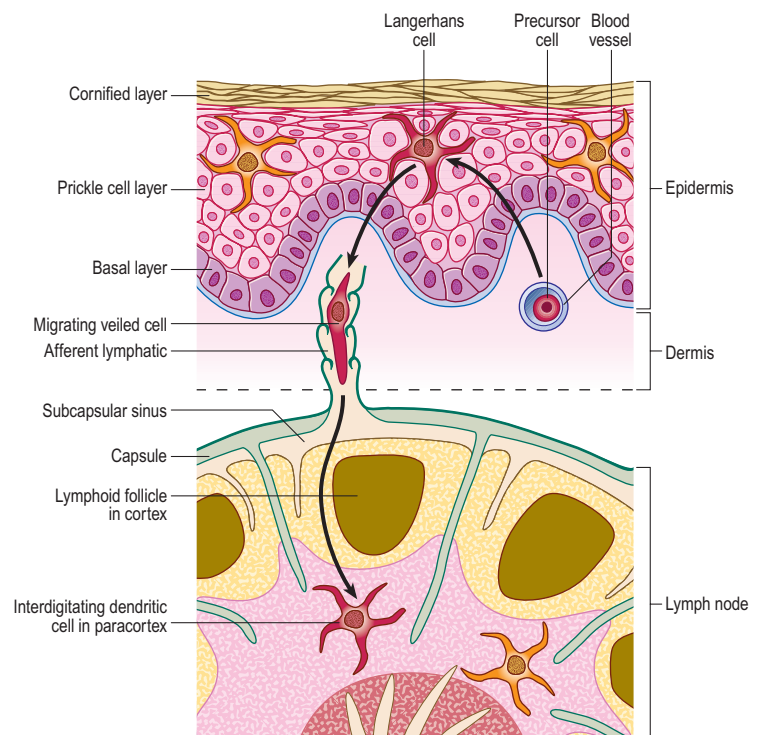


Fig. 4.14 Dendritic cells in the skin and lymphoid tissues. Their migratory routes are shown: from blood-borne, marrow-derived precursors to immature dendritic cells (Langerhans cells) in skin, and then to migrating veiled cells in afferent lymphatic vessels and interdigitating dendritic cells in lymph nodes. An example of each cell in the sequence (arrowed) is shown in red.

stratified squamous epithelia, e.g. the oral mucosa (Langerhans cells), and in the dermis and most other tissues (interstitial dendritic cells), where they are concerned with immune surveillance. Immature dendritic cells have an antigen-capturing function. They respond to chemotactic signals, e.g. defensins released by epithelial cells in the small intestine, and they express pattern recognition receptors (e.g. Toll-like receptors) on their surface. Binding of pathogen-associated molecular pattern molecules (PAMPs) derived from bacteria (e.g. carbohydrate, lipopolysaccharide or DNA) to these receptors stimulates the dendritic cells to become activated and migrate via the lymphatics to nearby secondary lymphoid tissues, where they can present antigen to T cells. They can also be activated by recognition of damage-associated pattern molecules (DAMPs), such as ATP, DNA, heat-shock proteins and high-mobility group box 1 (HMGB1) released from injured or necrotic cells. Mature dendritic cells are known as veiled cells when in the afferent lymphatics and the subcapsular sinuses of lymph nodes, and as interdigitating dendritic cells once they are within the lymphoid tissue proper. Their function within the secondary lymphoid tissue is to present their processed antigen to T lymphocytes, and thus to initiate and stimulate the immune response. For a review of research on dendritic cell function, see [Colonna et al \(2006\)](#).

Langerhans cells

Langerhans cells (see [Fig. 4.14](#)) are one of the best-studied types of immature dendritic cell (reviewed in [Berger et al \(2006\)](#), [Chopin and Nutt \(2014\)](#)). They are present throughout the epidermis of skin, where they were first described, but are most clearly identifiable in the stratum spinosum. They have an irregular nucleus and a clear cytoplasm, and contain characteristic elongated membranous vesicles (Birbeck granules). Langerhans cells endocytose and process antigens, undergoing a process of maturation from antigen-capturing to antigen-presenting cells that express high levels of MHC class I and II molecules, co-stimulatory molecules and adhesion molecules. They migrate to lymph nodes to activate T lymphocytes.

Interdigitating dendritic cells

Immature dendritic cells are found all over the body, including peripheral blood, and function in antigen-processing and immune surveillance. Mature dendritic cells are present in T-cell-rich areas of secondary lymphoid tissue (paracortical areas of lymph nodes, interfollicular areas of MALT, peri-arteriolar sheaths of splenic white pulp), where they are frequently referred to as interdigitating dendritic cells. Within the secondary lymphoid tissues, they are involved in the presentation to T lymphocytes of antigens associated with either MHC class I (CD8 T cells) or MHC class II (CD4 T cells) molecules. Naïve T cells can only

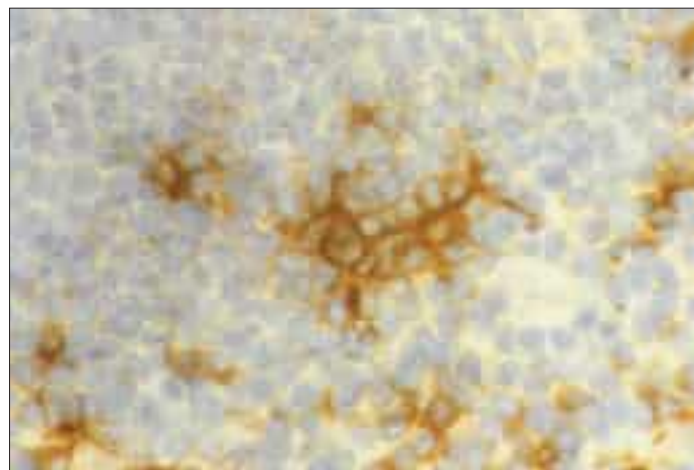


Fig. 4.15 Follicular dendritic cells (brown) in a germinal centre of the human palatine tonsil (immunoperoxidase-labelled). (Courtesy of Marta Perry MD, UMDS, London.)

respond to antigen presented by dendritic cells. The T cells are stimulated not only by recognition of the antigen–MHC complex by the TcR, but also by interaction with co-stimulatory molecules expressed by the dendritic cells, and by cytokines secreted by the cells. These cytokines not only help activate the T cell but can also direct the nature of the T-cell response (e.g. Th1 or Th2). Appropriate T cells are thus activated to proliferate and are primed for carrying out their immunological functions. Once primed, T cells can then be stimulated by any APC, including macrophages and B cells.

Follicular dendritic cells

Follicular dendritic cells (FDCs; [Fig. 4.15](#)) are a non-migratory population of cells found in the follicles of secondary lymphoid tissues, where they attract and interact with B cells. Unlike other dendritic cells, FDCs are not haemopoietic in origin but are probably derived from the stromal cells of lymphoid tissues. They are unable to endocytose and process antigen, and they lack MHC class II molecules. However, Fc receptors and complement receptors CD21 and CD35 on FDCs allow the cells to bind immune complexes to their surface for subsequent presentation, as unprocessed antigen, to germinal centre B cells. Interactions between B cells, CD4 helper T cells and FDCs in the germinal centres are important in the selection of high-affinity B cells and their maturation to either plasma cells or memory B lymphocytes ([Liu 2001](#)).

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